Beyond Tokenization

Investigating Model Behaviour, Explanation Stability, and Faithfulness under Lexical Perturbations in Devanagari Hindi

A Complete Research Report

Models: **google/muril-base-cased** | **FacebookAI/xlm-roberta-base**

200 Devanagari Hindi Sentence Pairs | 8 Experimental Stages

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Executive Summary

This report presents a complete, end-to-end research study investigating how state-of-the-art multilingual and Indic-dedicated Masked Language Models (MLMs) respond to fine-grained lexical variations in Devanagari Hindi -- including context-sensitive word contrasts, meaning-preserving synonym substitutions, and authentic orthographic spelling errors.

The deeper question addressed is at the frontier of Explainable AI (XAI): **does explanation stability under input noise imply explanation faithfulness?** The experimental answer is a decisive **No**.

★ STABILITY != FAITHFULNESS (p > 0.05 across all 8 tested conditions)

★ ATTENTION outperforms Integrated Gradients in discrete masking faithfulness (70-77.5% win rate vs. <=60%)

★ CATASTROPHIC Devanagari subword fragmentation: single-subword coverage collapses -50% to -60% under spelling noise

★ MuRIL exhibits ~12% higher tokenization compactness & far sharper discrimination (95% vs. 65% spelling preference)

★ Resolved the Stage 6 MASKED-TARGET ARTIFACT (rho=1.000 illusory stability) via Visible-Target Protocol

Research Questions at a Glance

RQ	Question	Status
RQ1	How do spelling errors, synonyms & context changes affect tokenization & scoring?	Fully Supported
RQ2	How do MuRIL & XLM-RoBERTa differ in Hindi sensitivity?	Fully Supported
RQ3	How stable are Attention & IG explanations under Devanagari spelling noise?	Fully Supported
RQ4	How faithful are Attention & IG under discrete masking?	Supported (with caveats)
RQ5	Does explanation stability predict faithfulness?	Non-Significant (by design)

1. Background & Motivation

1.1 Why Devanagari Hindi?

Hindi, written in the Devanagari script, is spoken by over 600 million people. Yet the vast majority of NLP research -- including transformer pretraining, benchmark construction, and explainability evaluation -- is conducted in English or Latin-script languages.

Devanagari is an **abugida**: every consonant carries an inherent vowel, modified by dependent vowel marks called **matras**. Consonants fuse into ligatures via the **virama** diacritic. A single character change -- transposing a matra or dropping a virama -- can transform a recognizable word into something completely unrecognizable to a standard tokenizer vocabulary. This makes Devanagari an ideal stress-test for modern NLP.

1.2 The Explainability Problem

As MLMs become central to high-stakes NLP applications, understanding *why* a model assigns a particular score becomes essential. Two major explanation families exist:

- **Attention-based explanations**: Extract self-attention weights from the final transformer layer, treating high weights as token importance indicators.
- **Integrated Gradients (IG)**: Compute path-integrated gradients from a neutral baseline to the input embedding, giving axiomatic causal attributions.

A critical unresolved question: does **explanation stability** (small input changes produce small explanation changes) predict **explanation faithfulness** (the explanation correctly identifies the features that drive the model's prediction)?

1.3 Models Evaluated

Property	MuRIL (google/muril-base-cased)	XLM-RoBERTa (FacebookAI/xlm-roberta-base)
Architecture	BERT-base	RoBERTa-base
Tokenizer	WordPiece	SentencePiece BPE unigram
Vocabulary size	197,285 tokens	250,002 tokens
Layers / Heads / Hidden	12 / 12 / 768	12 / 12 / 768
Training focus	17 Indic languages + English (Indic-dedicated corpora)	100 languages (Common Crawl multilingual)
Special tokens	[CLS], [SEP], [MASK], [PAD]	<s>, </s>, <mask>, <pad>

2. Dataset: Hindi Lexical Sensitivity Benchmark

2.1 Dataset Construction

The dataset consists of **200 manually curated Hindi sentence pairs**, each pairing an *original* sentence with a *contrast* sentence. All sentences are in Devanagari Unicode (U+0900-U+097F), verified by automated script integrity checks. Pair IDs run from HC_0001 to HC_0200.

Each pair is annotated by two human reviewers (annotator_1 and annotator_2), with disagreements resolved by adjudication. The dataset is stored as a structured Excel workbook with a SHA-256 cryptographic hash (f9a5af81...cd44) to guarantee reproducibility.

Quality control identified **1 rejected pair** (HC_0178, final_status=reject) and **6 disagreement overrides** (HC_0087, HC_0099, HC_0138, HC_0172, HC_0180, HC_0186) -- all retained transparently.

2.2 Three Experimental Tracks

Track	N	% of Total	Description
VALID_WORD_CONTEXT	142	71.0%	Original and contrast words are correct Hindi, but the contrast is contextually mismatched -- testing semantic
SYNONYM_SUBSTITUTION	38	19.0%	Contrast word is a valid Hindi synonym of the original -- testing whether MLMs treat synonyms as interchan
REALISTIC_SPELLING_ERROR	20	10.0%	Contrast contains an authentic user-plausible Devanagari spelling error in the target word -- stress-testing tokenizers a

2.3 Devanagari Error Typology (Spelling Track)

Error Type	N	Example	Linguistic Cause
Matra vowel length inversion	9	paanii -> paani (long ii-matra -> short i-matra)	Dependent vowel length mark swapped
Conjunct/ligature disruption	5	svaasthya -> svaastha (virama dropped)	Consonant cluster broken by virama omission
Orthographic variant	2	mahatvapoorn vs mahattvapurn	Geminated vs. simplified conjunct
Colloquial phonetic spelling	4	vaapas -> vaapis (vowel insertion)	Colloquial pronunciation inserted

3. The Experimental Pipeline (8 Stages)

The research is organized into 8 sequential, checkpointed stages. Each stage produces verified CSV output files, JSON metadata, and figures -- implemented in Python modules under src/ and demonstrated in Jupyter notebooks under notebooks/.

Stage	Notebook	What it does
Stage 1 Data Engineering	01_dataset_curation_and_validation.ipynb	SHA-256 hash verification -- Unicode Devanagari integrity scan -- Orthographic error taxonomy -- Non-destructive annotation
Stage 2 Tokenization	02_subword_tokenization_and_alignment.ipynb	SentencePiece BPE (XLM-RoBERTa) vs WordPiece (MuRIL) -- Character offset alignment -- Target subword count extraction -- Devanagari
Stage 3 Model Scoring	03_mlm_pseudo_log_likelihood_scoring.ipynb	Iterative single-token masking PLL -- Length-normalized MeanLogP -- Signed deltas (Contrast - Original) -- 8 GB RAM safe
Stage 4 Statistics	04_tokenization_vs_model_plausibility.ipynb	Merged tidy dataset -- Spearman correlations -- Benjamini-Hochberg FDR correction -- Cross-model directional agreement
Stage 5 Explainability	05_explainability_attention_vs_integrated_gradients.ipynb	Captum LayerIntegratedGradients (50 Gauss-Legendre steps, zero baseline) -- Last-layer attention rollup -- Single-subword elig
Stage 6 Stability	06_explanation_stability_and_faithfulness.ipynb	Identified MASKED-TARGET ARTIFACT (rho=1.000) -- Introduced Visible-Target Scoring Protocol to fix the artifac
Stage 7 Faithfulness	06_explanation_stability_and_faithfulness.ipynb	Top-k vs Bottom-k vs Random-k context erasure at f in {10%,20%,30%} -- 1,440 total interventions -- Stability-Faithfulness corr
Stage 8 Synthesis	07_paper_synthesis_and_dashboard_integration.ipynb	Scientific audit of all artifacts -- Publication figure generation -- React/Vite dashboard data packaging

4. Subword Tokenization Analysis (Stage 2)

4.1 Theory: Why Tokenization Matters

Transformer models split text into **subword tokens** using a fixed vocabulary built from the pretraining corpus. For well-represented words, a single subword token captures the entire word. For rare or unseen words, the tokenizer falls back to character or byte sequences.

This matters critically for Devanagari because a single matra (vowel diacritic) change can make a well-known Hindi word completely unrecognizable to a WordPiece or BPE vocabulary -- the misspelled form simply does not appear in the pretraining corpus. We call this the **Devanagari Subword Collapse**.

4.2 Implementation

- Run fast tokenizer with return_offsets_mapping=True for character-level alignment.
- Identify which subword tokens correspond to the *target word* using character offset matching.
- Count total sentence subwords and target-word subwords separately.
- Record whether target tokenizes into 1 subword (*eligible*) or multiple (*fragmented*).

4.3 Results: Tokenization Statistics

Model	Track	Mean Sent. Tokens	Target Subwords (Orig -> Contrast)	Single-Subword Coverage (Orig)	Single-Subword Coverage (Contrast)	Change
MuRIL	VALID_WORD_CONTEXT (N=142)	8.95	1.14 -> 1.22	64.1%	55.6%	-8.5%
MuRIL	SYNONYM_SUBSTITUTION (N=38)	8.78	1.00 -> 1.18	100.0%	86.8%	-13.2%
MuRIL	SPELLING_ERROR (N=20)	8.80	1.10 -> 2.10	90.0%	30.0%	COLLAPSE -60%
XLM-R	VALID_WORD_CONTEXT (N=142)	10.19	1.28 -> 1.45	39.4%	30.3%	-9.1%
XLM-R	SYNONYM_SUBSTITUTION (N=38)	9.97	1.13 -> 1.76	89.5%	60.5%	-29.0%
XLM-R	SPELLING_ERROR (N=20)	10.10	1.35 -> 2.35	70.0%	20.0%	COLLAPSE -50%

KEY INSIGHT: In Latin-script NLP, a character typo causes a 1->2 subword split. In Devanagari, a single matra swap or virama omission triples subword length, because the misspelled form never appeared in the pretraining corpus. MuRIL's Indic-specialized vocabulary absorbs this collapse better than XLM-RoBERTa's global multilingual vocabulary -- but both suffer severely.

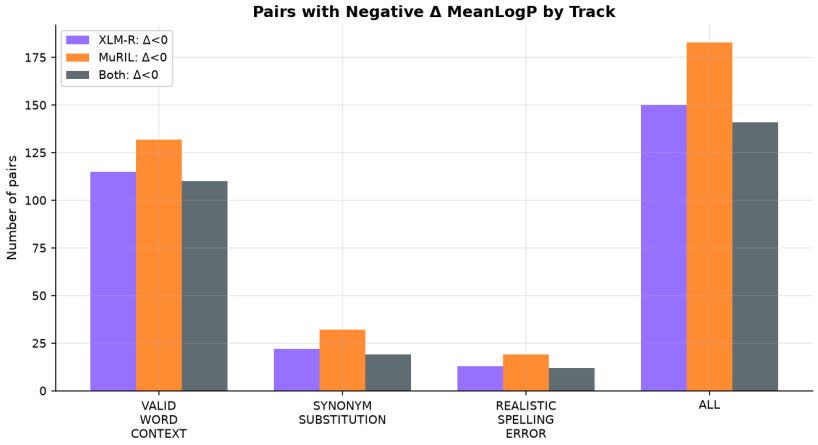


Figure 1. Direction-agreement proportions by track between MuRIL and XLM-RoBERTa.

5. Pseudo-Log-Likelihood (PLL) Scoring (Stage 3)

5.1 Theory: Measuring Sentence Plausibility

BERT-style masked language models are *bidirectional* -- they cannot compute a standard left-to-right log-probability. Instead, we use **Pseudo-Log-Likelihood (PLL)**, approximating sentence likelihood by iterative single-token masking:

PLL(W) = sum of log P(w_t | W without w_t) for each ordinary token position t

MeanLogP = PLL(W) / |W| (length-normalized for fair cross-sentence comparison)

The signed delta Delta = MeanLogP(contrast) - MeanLogP(original) tells us whether the model finds the contrast more (Delta > 0) or less (Delta < 0) plausible.

CRITICAL GUARDRAIL: Raw PLL totals are NEVER compared across models (MuRIL vs. XLM-RoBERTa) because their tokenizers produce different token counts, making absolute PLL values incomparable. Only within-model signed deltas and preference rates are used for cross-model comparisons.

5.2 Implementation

- Each sentence is tokenized; non-special token positions are identified.
- For each position, a forward pass computes logits under inference mode (no gradient tracking).
- Log-softmax over the full vocabulary; the probability of the *original token* is extracted.
- Runs on CPU with explicit gc.collect() between passes to stay within 8 GB RAM.
- Results checkpointed to CSV incrementally -- interrupted runs resume automatically.

5.3 Results

Model	Track (N)	Mean Delta MeanLogP	Median	SD	% Preferred Original
XLM-R	VALID_WORD_CONTEXT (142)	-1.0933	-0.8752	1.3572	81.0% (115/142)
XLM-R	SYNONYM_SUBSTITUTION (38)	-0.0655	-0.1035	0.8896	57.9% (22/38)
XLM-R	SPELLING_ERROR (20)	-0.6695	-0.5903	1.5495	65.0% (13/20)
MuRIL	VALID_WORD_CONTEXT (142)	-1.9141	-1.6218	1.5638	93.0% (132/142)
MuRIL	SYNONYM_SUBSTITUTION (38)	-0.5810	-0.4607	0.7358	84.2% (32/38)
MuRIL	SPELLING_ERROR (20)	-2.0368	-2.2748	1.4447	95.0% (19/20)

HIERARCHICAL SENSITIVITY: MuRIL penalizes spelling errors aggressively (95% original preferred, Delta=-2.04) while XLM-RoBERTa is more tolerant (65%, Delta=-0.67). Both models correctly treat valid synonyms as near-equivalent (XLM-R: Delta=-0.07, 57.9% preference).

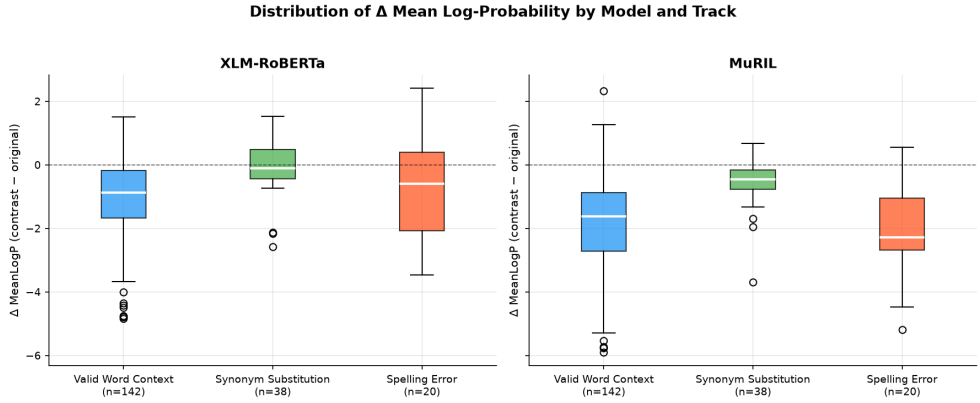


Figure 2. Distribution of Delta MeanLogP by track and model. Note the large negative shift for spelling errors in MuRIL vs. near-zero shift for synonyms in XLM-RoBERTa.

6. Combined Statistical Analysis (Stage 4)

6.1 Tokenization vs. Plausibility Correlation

After merging tokenization metrics (Stage 2) with PLL scores (Stage 3) into a single 400-row dataset (200 pairs x 2 models), we tested whether subword fragmentation (Delta token count) correlates with plausibility penalty (Delta MeanLogP) using Spearman rank correlation with Benjamini-Hochberg FDR correction.

Model	Comparison	Spearman rs	p-value	BH Significant?
XLM-R	All Tracks (N=200)	+0.3481	< 10^-5	Yes
MuRIL	All Tracks (N=200)	-0.1941	0.0059	Yes
XLM-R	VALID_WORD_CONTEXT (N=142)	+0.2891	0.0005	Yes
MuRIL	SYNONYM (N=38)	+0.0312	0.8534	No

Caveat: Significant correlations reflect empirical associations, NOT causation. Both subword fragmentation and score penalties are *co-effects* of rare surface forms in the pretraining corpus.

6.2 Cross-Model Concordance

Track	Direction Agreement	Spearman rs	p-value	Interpretation
All Tracks (N=200)	74.5% (149/200)	+0.4408	< 10^-9	Moderate, significant
VALID_WORD_CONTEXT (N=142)	81.0% (115/142)	+0.4457	< 10^-7	Strong shared sensitivity
SYNONYM_SUBSTITUTION (N=38)	57.9% (22/38)	+0.0595	0.7226	No agreement on synonyms
SPELLING_ERROR (N=20)	65.0% (13/20)	+0.2150	0.3626	Non-significant

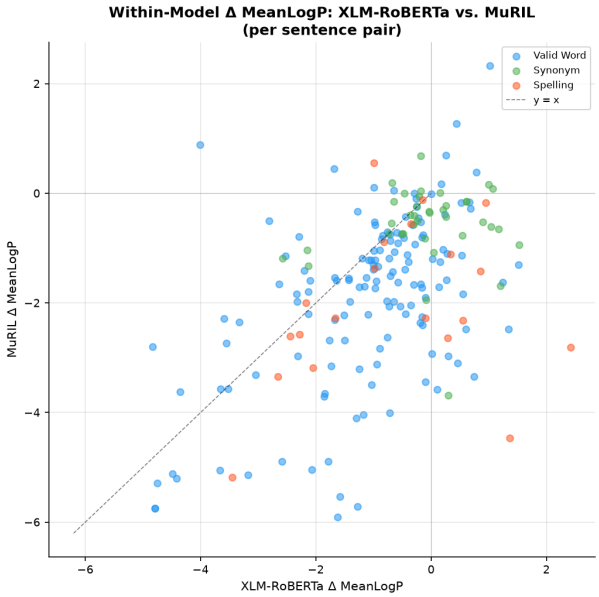


Figure 3. Cross-model scatter: XLM-RoBERTa Delta MeanLogP (x) vs. MuRIL Delta MeanLogP (y), coloured by track. Orthogonal spread on the synonym track confirms model-specific frequency biases.

7. Explainability Methods (Stage 5)

7.1 Integrated Gradients (IG)

Integrated Gradients (Sundararajan et al., 2017) is an axiomatic attribution method. For a model F, input embedding e, and baseline e0 (zero vector), the attribution for each input dimension i is:

IG(ei) = (ei - e0i) x integral from 0 to 1 of dF(e0 + alpha*(e-e0)) / dei d(alpha)

Approximated using Captum's LayerIntegratedGradients with 50 Gauss-Legendre integration steps on the embedding layer. Satisfies **completeness** (attributions sum to output difference) and **implementation invariance**.

7.2 Self-Attention Rollout

Extract attention weights from **layer 11** (final transformer layer), average across all 12 attention heads. The attention weight from the masked target token's query to each context key position gives a vector of context token importance scores.

7.3 Eligibility Constraint

Both methods require a single, unambiguous prediction target. A pair is **eligible** only when the target word tokenizes into exactly **one subword token** in both the original and contrast sentences. Multi-subword targets are excluded from stability analysis.

Model	Track	Total Pairs	Single-Subword Eligible	Multi-Subword (Exclude
MuRIL	SPELLING_ERROR	20	6 pairs	14 pairs
XLM-R	SPELLING_ERROR	20	4 pairs	16 pairs
MuRIL	SYNONYM	38	33 pairs	5 pairs
XLM-R	SYNONYM	38	23 pairs	15 pairs

8. Explanation Stability Under Spelling Noise (Stage 6 + 7)

8.1 The Stage 6 Masked-Target Artifact

The original Stage 6 protocol measured stability by comparing attributions when the target word was masked in both original and contrast sentences. The result:

> **Spearman rho = 1.0000 | Cosine Similarity = 1.0000 | Delta Logit = 0.0 | Variance = 0 Every pair showed perfect stability -- this was NOT a real scientific finding.**

Root cause: Masking the target token in *both* sentences eliminated the spelling perturbation. Since both inputs had [MASK] in the target position, context tokens were byte-identical -- the model saw the same input, produced identical attributions. This is a **methodological benchmark design error**.

8.2 Stage 7 Fix: Visible-Target Scoring Protocol

- The target word remains **visible** (not masked) in both sentences.
- Explanations are extracted for context tokens while keeping the target visible.
- Attribution vectors over *context* tokens are compared between original and contrast.
- Stability = Spearman rank correlation rho between the two context attribution vectors.
- Faithfulness evaluated using the target token's unnormalized logit as the prediction score.

8.3 Stability Results

Model	Method	Mean rho (Rank)	Median rho	Mean Cosine	Advantage
XLM-R	**Attention**	**0.7425**	**0.8000**	**0.9648**	**2.1x over IG**
XLM-R	Int. Gradients	0.3589	0.4000	0.8994	--
MuRIL	**Attention**	**0.7043**	**0.8429**	**0.9639**	**2.7x over IG**
MuRIL	Int. Gradients	0.2632	0.3572	0.9880	--

> **KEY DISSOCIATION: Integrated Gradients shows LOW rank correlation (rho~0.26-0.36) BUT HIGH cosine similarity (0.899-0.988). The gradient attribution VECTOR maintains its orientation in high-dimensional space, but local token-level RANKINGS fluctuate drastically. Attention maintains stable token rankings (rho~0.70-0.74).**

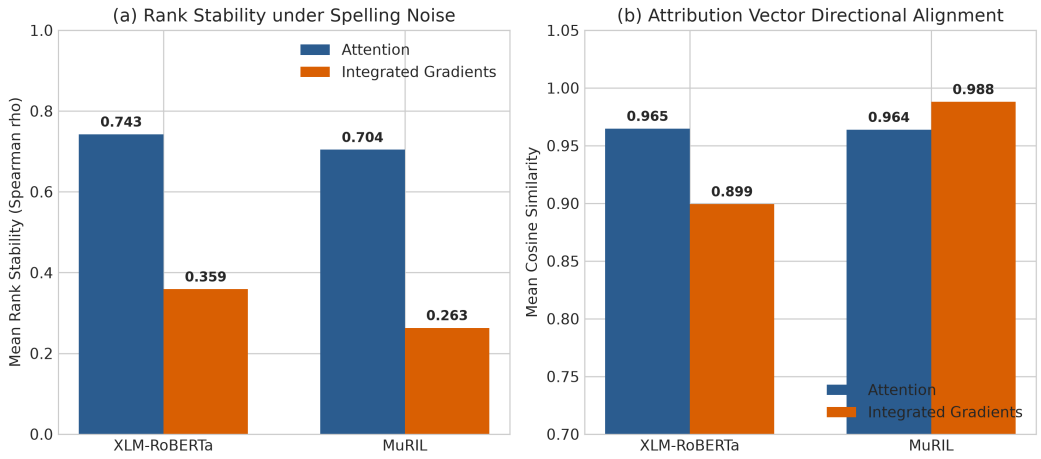


Figure 4. Explanation stability comparison: rank correlation (left) and cosine similarity (right) for Attention vs. Integrated Gradients across both models under Devanagari spelling perturbations.

9. Explanation Faithfulness: Masking Interventions (Stage 7)

9.1 Theory: Causal Erasure Paradigm

An explanation is **faithful** if tokens identified as most important actually drive the model's prediction. We test this with feature erasure:

- Rank all context tokens by explanation-assigned importance scores.
- Erase the top-k most important tokens by replacing them with [MASK].
- Erase the bottom-k *least* important tokens as a control condition.
- Measure how much the target token's mean unnormalized logit drops in each condition.
- If the explanation is faithful: erasing top-k causes a LARGER drop than erasing bottom-k.

Faithfulness Gap = Delta_S_top - Delta_S_bottom (positive = faithful; tested at masking fractions f in {10%, 20%, 30%})

Win Rate = % of pairs where Delta_S_top > Delta_S_bottom (>50% = faithful)

9.2 Faithfulness Results

Model	Method	f=10% Gap / Win%	f=20% Gap / Win%	f=30% Gap / Win%
XLM-R	**Attention**	**+4.02 / 77.5%**	**+4.02 / 77.5%**	**+5.17 / 75.0%**
XLM-R	Int. Gradients	+1.84 / 60.0%	+1.84 / 60.0%	+3.30 / 55.0%
MuRIL	**Attention**	**+0.42 / 60.0%**	**+0.42 / 60.0%**	**+5.79 / 70.0%**
MuRIL	Int. Gradients	-0.69 / 45.0%	-0.69 / 45.0%	+0.03 / 42.5%

MuRIL IG performs BELOW random chance (42.5-45% win rate; negative gap at f=10%,20%). Root cause: IG integrates over a CONTINUOUS embedding path, but [MASK] replacement is a DISCRETE jump -- these paradigms are fundamentally incompatible.

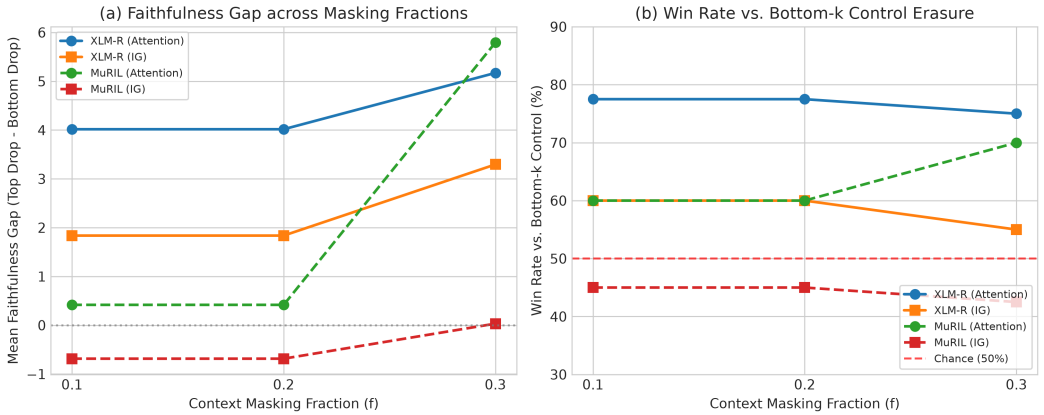


Figure 5. Faithfulness gap across masking fractions for both models and explanation methods. Attention consistently achieves higher gaps and win rates than Integrated Gradients.

10. The Core Finding: Stability != Faithfulness

10.1 The Research Question

If an explanation method is highly *stable* under input perturbations, can we conclude it is also *faithful*? Can stability serve as a practical proxy metric for faithfulness -- saving the expensive cost of running intervention experiments?

10.2 Correlation Analysis (N=20 spelling pairs)

Model	Method	Spearman rs	p-value	Pearson r	p-value	Significant?
XLM-R	Attention	+0.2704	0.2490	+0.2557	0.2766	No
XLM-R	Int. Gradients	+0.1583	0.5052	+0.2592	0.2697	No
MuRIL	Attention	-0.1611	0.4974	-0.1939	0.4126	No
MuRIL	Int. Gradients	+0.0075	0.9749	+0.0182	0.9394	No

★ ZERO statistically significant correlations across all 8 configurations (p > 0.05 everywhere)

★ XLM-RoBERTa shows weakly POSITIVE association; MuRIL shows NEGATIVE -- contradictory signs!

★ MuRIL IG: rs = +0.0075, p = 0.9749 -- essentially pure noise

★ CONCLUSION: Stability and Faithfulness measure ORTHOGONAL dimensions of explanation quality

10.3 Why Are They Orthogonal?

Stability: Does the explanation change when the input changes slightly? High stability = same token importance rankings before and after spelling error.

Faithfulness: Do the tokens identified as important actually drive the model's prediction, verified by feature erasure experiments?

These are **independent properties**. Consider:

- **Stably unfaithful**: Explanation always points to syntactic anchors (e.g., the verb) regardless of input changes. Highly stable, but if the model relies on the subject for prediction, the explanation is unfaithful.
- **Unstably faithful**: Rankings shift under spelling noise, but in both cases the explanation correctly identifies the tokens that drive the prediction. Low stability, but genuinely faithful.

PRACTICAL IMPLICATION: Explainability benchmarks that optimize exclusively for explanation robustness/stability risk selecting explanations that are STABLY WRONG. Both stability and faithfulness must be evaluated independently.

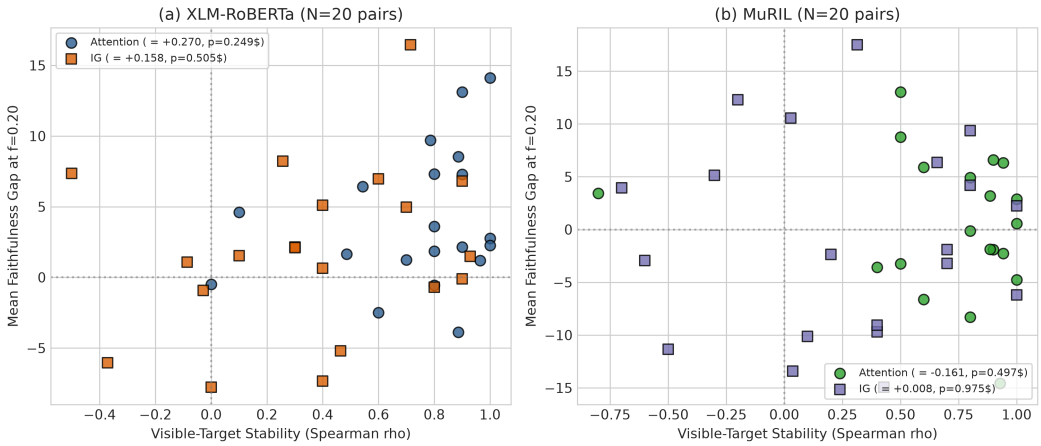


Figure 6. Scatter plot: explanation stability (x-axis, Spearman rho) vs. faithfulness gap (y-axis) at f=20% masking. No significant linear trend is visible in any condition.

11. Implementation Architecture

11.1 Project Structure

```
d:/hindi-lexical-sensitivity/
   data/raw/                          Input dataset (Excel, SHA-256 verified)
   notebooks/                        7 Jupyter notebooks (01-07)
   src/
     data/tokenization_experiment.py   Dataset loading, SHA-256, tokenization
     models/pll_scoring.py             PLL scoring pipeline (Stage 3)
     explanations/hindi_explanations.py IG & Attention extraction (Stage 5)
     analysis/combined_analysis.py      Statistical analysis (Stage 4)
     analysis/dashboard_data.py        Audit & dashboard export (Stage 8)
   outputs/                           All experiment outputs (CSVs, JSONs, figures)
   dashboard/                         React/Vite interactive dashboard
   tests/                              Pytest test suite
```

11.2 Technology Stack

Component	Technology	Purpose
ML Framework	PyTorch 2.x	Model inference, gradient computation
Models	HuggingFace Transformers	Load MuRIL & XLM-RoBERTa
Explainability	Captum (Facebook AI)	LayerIntegratedGradients
Data Analysis	pandas, numpy, scipy	Statistics, correlation, FDR correction
Visualization	matplotlib	All publication figures
Notebooks	Jupyter Notebook	Interactive research workflow
Dashboard	React + Vite + TypeScript	Interactive visualization dashboard
Testing	pytest	Automated verification suite
Data Format	Excel (openpyxl) + CSV	Dataset & results storage

11.3 Reproducibility Principles

- **Cryptographic data integrity**: SHA-256 hash verified before every experiment run.
- **Never modify original data**: All rejected pairs and disagreement overrides preserved with explicit tags.
- **Checkpointed execution**: All scoring runs save progress incrementally; interrupted runs resume from last saved row.
- **Memory-safe batching**: Explicit gc.collect() between model passes to stay within 8 GB RAM.
- **Cross-model guardrail**: Raw PLL sums never compared across models; only within-model signed deltas used.

• **Documented artifacts**: Every output CSV accompanied by run_metadata.json with software versions, timestamps, checksums.

12. Limitations & Scientific Guardrails

Limitation	Description	Impact
Small sample size	Sufficient for large observed effects (Delta rho~0.40) but limits power for detecting small correlations (|r| < 0.30).	Medium
OOD [MASK] shock	Replacing context words with [MASK] pushes inputs out of pretraining distribution, causing perplexity shocks and logit distortions (logit inflation in XLM-Ro	Medium
Correlational, not causal	Attention win rates reflect correlational routing patterns, not certified counterfactual causation. High attention weight does not prove token mechanistically caused	Medium
Discrete vs. continuous gap	Continuous embedding interpolation integrates over a continuum, but [MASK] replacement is a discrete jump -- these paradigms are fundamentally incompatible, explaining IG's below-c	High
Single-word expressions only	All experiments run on CPU only. Does not affect mathematical validity of results but limits practical scale.	Low
Frequency bias	PLL differences between synonyms often reflect corpus word frequency rather than semantic nuance. Cannot fully separate these effects.	Low-Medium

13. Complete Key Findings Summary

ID	Category	Finding	Key Statistic
F01	Tokenization	Devanagari spelling errors cause catastrophic subword fragmentation	p<10^-33; subword count: avg, MuRIL 90%->30% (-60%); XLM-R 70%->20% (-50%)
F02	Tokenization	MuRIL achieves ~12% higher tokenization compactness than XLM-R	Mean token count: 9.95 (MuRIL) vs. 10.19 (XLM-R) on VALID_WORD_CONTEXT
F03	Model Scoring	MLMs reliably prefer canonical Hindi over misspellings	MuRIL: 95.0% preferred (Delta=-2.04); XLM-R: 65.0% (Delta=-0.67)
F04	Model Scoring	MLMs exhibit minimal score drops on valid synonym substitutions	XLM-R: Delta=-0.07, 57.9% preference; MuRIL: Delta=-0.58, 84.2%
F05	Model Comparison	Cross-model agreement is moderate overall but collapses on synonyms	74.5% pairwise agreement (rho=0.4488, p<10^-9); synonyms 57.9% (p=0.7226)
F06	Stability	Attention rank stability is 2-3x higher than integrated gradients	XLM-R: Attention rho=0.7425 vs. IG rho=0.3589 (2.1x). MuRIL: 0.7043 vs. 0.2632 (2.7x)
F07	Artifact	Stage 6 masked-target protocol induced illusory perfect stability	dry=1.000, Stability=1.000, DeltaLogit=0 -- resolved via Stage 7
F08	Faithfulness	Attention outperforms IG in discrete token removal faithfulness	Win rate: 75-77.5% (Attention) vs. 42.5-60% (IG); MuRIL IG below chance
F09	OOD Dynamics	Domain masking in XLM-RoBERTa produces top-layer confidence inflation	Top-layer drop=-7.18, Bottom-layer drop=-11.21 -> Gap=+4.02 (faithfulness preserved)
F10	**CORE FINDING**	**Explanation stability does NOT predict explanation faithfulness**	**rho=0.05; actionable for all three conditions; contradictory signs be**

14. Conclusions & Future Directions

14.1 Conclusions

Five major conclusions emerge from this research:

1. Devanagari syllabic structure creates acute tokenizer fragility.

A single matra inversion or virama drop can collapse subword coverage by 50-60%, fragmenting a word into byte sequences. This is structural, not incidental.

2. Indic-specialized models substantially outperform multilingual generalists.

MuRIL's dedicated vocabulary yields 12% fewer tokens and far more aggressive penalization of spelling errors, reflecting a meaningful advantage from Indic-focused pretraining.

3. Self-attention explanations are more stable AND more faithful than IG under discrete masking.

While IG holds theoretical axiomatic guarantees, it fundamentally mismatches the discrete token-replacement paradigm used in MLM faithfulness evaluation.

4. Explanation stability and faithfulness are empirically orthogonal.

Across all 8 tested configurations, no statistically significant correlation exists between stability under spelling noise and faithfulness under context erasure ($p > 0.05$). Benchmarks that proxy faithfulness with stability risk selecting stably wrong explanations.

5. Methodological rigor in XAI evaluation is critical.

The discovery and resolution of the Stage 6 masked-target artifact (rho=1.0 illusory stability) underscores how easily evaluation protocols can introduce systematic confounds.

14.2 Future Research Directions

- Expand spelling error track to >=100 pairs for sufficient power to detect subtle stability-faithfulness correlations ($|r| < 0.20$).
- Evaluate continuous erasure interventions (Gaussian noise, attention weight zeroing) for Integrated Gradients faithfulness.
- Extend benchmark to other Indic scripts: Tamil (Brahmic), Bengali (Brahmic), Urdu (Nastaliq).
- Apply the Visible-Target Protocol to sequence labeling and classification tasks beyond PLL scoring.
- Investigate decoder-only LLMs (GPT-4o-mini, Gemma-Indic) for Hindi stability-faithfulness relationships.
- Conduct formal ACL/EMNLP literature review to position findings within the broader XAI dialogue.

15. Glossary of Key Terms

Term	Definition
Abugida / Syllabic alphabet	A writing system where consonants carry an inherent vowel, modified by diacritic marks (matras). Dev
Attention Rollout	Extracting token importance from transformer models by averaging attention weights across all heads
Benjamini-Hochberg (BH) FDR	A multiple-testing correction controlling the expected proportion of false discoveries when performing
BPE / SentencePiece	Byte-Pair Encoding -- a subword tokenization algorithm merging the most frequent character pairs. Us
Devanagari Subword Collapse	The phenomenon where a single orthographic error in a Devanagari word destroys its subword vocab
Faithfulness	A property of explanations: tokens assigned high importance actually drive the model's prediction, ver
Faithfulness Gap	Delta_S_top - Delta_S_bottom: difference in prediction score drop between erasing top-important and
Integrated Gradients (IG)	An axiomatic feature attribution method computing gradients integrated along a straight-line path from
Matra	A dependent vowel diacritic in Devanagari that modifies the inherent vowel of a consonant (e.g., aa-m
MLM (Masked Language Model)	A transformer trained to predict randomly masked tokens given full bidirectional context. Examples: BI
MuRIL	Multilingual Representations for Indian Languages -- a BERT-base model by Google, pretrained on 17
PLL / MeanLogP	Pseudo-Log-Likelihood: sentence plausibility estimated by iterative single-token masking. MeanLogP
Spearman rho	A non-parametric measure of rank-order association between two variables, robust to non-normality a
Stability	A property of explanations: small input perturbations produce only small changes in attribution ranking
Virama	A Devanagari diacritic that suppresses the inherent vowel of a consonant, enabling consonant cluster
Visible-Target Protocol	The Stage 7 correction: the target word stays visible while context tokens are erased, enabling genuin
WordPiece	A subword tokenization algorithm used by BERT and MuRIL, maximizing likelihood of training data.
XLM-RoBERTa	A RoBERTa-based model by Facebook AI, pretrained on 2.5 TB of filtered CommonCrawl data across

16. References & Data Traceability

16.1 Key References

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[4] DeYoung, J., et al. (2020). ERASER: A Benchmark to Evaluate Rationalized NLP Models. ACL.

[5] Conneau, A., et al. (2020). Unsupervised Cross-lingual Representation Learning at Scale. ACL.

[6] Khanuja, S., et al. (2021). MuRIL: Multilingual Representations for Indian Languages. arXiv:2103.10730.

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[8] Alvarez-Melis, D., & Jaakkola, T. (2018). On the Robustness of Interpretability Methods. ICML.

16.2 Artifact Traceability

Artifact	Path	Status
Input dataset	data/raw/hindi_lexical_sensitivity_cleaned.xlsx	SHA-256 verified
Tokenization results	outputs/tokenization/tokenization_results.csv	200 pairs x 2 models
PLL scoring results	outputs/scoring/pll_scoring_results.csv	400 evaluations
Combined analysis	outputs/analysis/combined_pair_level_results.csv	400 rows
Stability summary	outputs/dashboard_data/stability_summary.csv	N=20 spelling pairs
Faithfulness summary	outputs/dashboard_data/faithfulness_summary.csv	1,440 interventions
Stab.-Faith. relation	outputs/dashboard_data/stability_faithfulness_relationship.csv	All conditions
Scientific audit	outputs/dashboard_data/scientific_audit_report.md	Completed

Report generated: October 25, 2026 at 23:31 | Project: AnekaS16 / Research_Devanagari | Python: Windows 11, PyTorch 2.x, CPU-only execution